Tab 3 — ROIC vs WACC Analysis (the 'Moat' tab)

Visa Inc. · $ in millions unless noted · market inputs at the Oct 14, 2026 close · blue = input · black = formula · green = link · yellow = judgement call

WACC inputs — Metric	Value	Source / formula
Cost of equity (CAPM)		
Risk-free rate (10-year U.S. Treasury par yield)	5.19%	*U.S. Treasury Daily Par Yield Curve, 10 Yr, Oct 14, 2026 close (latest official print): 5.19% — https://home.treasury.gov/resource-center/data-chart-center/interest-rates/TextView?type=daily_treasury_yield_curve&field_tdr_date_value_month=202610*
Visa raw beta (5-year, monthly vs S&P 500)	0.77	*Yahoo Finance 'Beta (5Y Monthly)' 0.77 (CNBC shows 0.82) — third-party figures; no primary source publishes beta. The sensitivity grid covers 0.59–1.10*
Blume weight on raw beta	0.67	*Standard Blume adjustment (Bloomberg convention): adjusted β = 0.67 × raw + 0.33 × 1.0*
Adjusted beta (used)	**0.85**	*Formula: Blume-adjusted beta — pulls a low raw beta toward 1.0*
Equity risk premium	5.50%	*Base case: 5.5% (conservative). Damodaran's implied ERP is 4.20% (Oct 1, 2026) — see memo below the WACC*
Cost of equity (Ke)	**9.84%**	*Formula: Rf + β × ERP*
Cost of debt		
Visa 4.150% senior notes due Dec 14, 2035 — yield (CUSIP 92826CAE2)	5.42%	*Public.com quote: price 90.95, yield 5.42% (no timestamp; may be yield-to-worst given Jun 2035 par call). Terms: SEC FWP https://www.sec.gov/Archives/edgar/data/1403161/000119312515399417/d102656dfwp.htm · undated quote; FINRA TRACE (CUSIP 92826CAE2) is the reference for trade-level data*
Pre-tax cost of debt (market)	5.42%	*Formula: = Visa's traded ~9-year USD bond yield (a marginal cost, not the ~3% book rate)*
Tax rate	17.10%	*Link: Tab 2 FY25 effective tax rate*
After-tax cost of debt (Kd)	**4.49%**	*Formula: Pre-tax Kd × (1 – tax rate)*
Memo: interest expense, 9 months to Jun 30, 2026	566	*Q3 FY26 earnings release (8-K) — https://www.sec.gov/Archives/edgar/data/0001403161/000140316126000103/q32026earningsrelease.htm*
Capital structure (market values)		
Share price ($) — close Oct 14, 2026	$391.43	*Yahoo Finance historical close (Google Finance prior close $391.46 agrees). Same date as the Treasury yield → one consistent valuation date*
As-converted class A shares outstanding (millions)	1,880	*Q3 FY26 Form 10-Q, Note 11 Stockholders' Equity, as-converted table at Jun 30, 2026: class A 1,702 + B-1 3 + B-2 1 + B-3 91 + C 73 + preferred A/B/C 10 = 1,880 — https://www.sec.gov/Archives/edgar/data/1403161/000140316126000104/R20.htm*
Market capitalisation	**735,888**	*Formula: Price × as-converted shares*
Total debt (Jun 30, 2026)	23,858	*Q3 FY26 release: current maturities $2,996M + long-term $20,862M · book value as proxy for market value*
Memo: effective interest rate on existing debt	3.16%	*Formula: annualised 9M interest / debt · legacy coupons (2.0–3.9%) are below today's market cost*
Equity weight	96.9%	*Formula: E / (D + E)*
Debt weight	3.1%	*Formula: 1 – equity weight*
WACC	**9.67%**	*Formula: E-weight × Ke + D-weight × Kd(after-tax)*
Memo: implied credit spread (bond yield – Rf)	0.23%	*About 0.2%: tight but plausible for an AA- issuer with ~9 years to maturity*
Memo: Damodaran implied ERP (Oct 1, 2026)	4.20%	*Damodaran Online home page: trailing-12-month, adjusted payout; prior month 4.14% — https://pages.stern.nyu.edu/~adamodar/New_Home_Page/home.htm*
Memo: WACC at Damodaran implied ERP	8.61%	*Formula: same WACC with the market-implied ERP — the 'less conservative' case*

Line item	FY21	FY22	FY23	FY24	FY25	FY26P	FY27P	FY28P	FY29P	FY30P	Formula / note
ROIC — including goodwill (return on everything shareholders paid, incl. Visa Europe)											
EBIT ex-litigation	15,807	19,681	21,927	24,057	26,556	30,104	33,355	36,899	40,491	44,323	*Link: Tab 2*
Normalised tax rate	17.0%	17.5%	17.9%	17.4%	17.1%	16.0%	17.0%	17.0%	17.0%	17.0%	*Link: Tab 2 (FY21 normalised to 17.0%)*
NOPAT	**13,120**	**16,231**	**18,004**	**19,859**	**22,016**	**25,287**	**27,685**	**30,626**	**33,608**	**36,788**	*Formula: EBIT × (1 – tax rate)*
Invested capital — ending	38,349	37,373	37,147	42,253	43,084	46,292	47,092	47,892	48,692	49,492	*Link: Tab 2 (Debt + Equity – Cash)*
Invested capital — average		37,861	37,260	39,700	42,669	44,688	46,692	47,492	48,292	49,092	*Link: Tab 2 · FY21 blank (needs FY20 balance sheet)*
ROIC — on ending capital	34.2%	43.4%	48.5%	47.0%	51.1%	54.6%	58.8%	63.9%	69.0%	74.3%	*Formula: NOPAT / ending invested capital*
ROIC — on average capital (primary)		**42.9%**	**48.3%**	**50.0%**	**51.6%**	**56.6%**	**59.3%**	**64.5%**	**69.6%**	**74.9%**	*Formula: NOPAT / average invested capital · preferred: matches a full year of profit to the capital employed through the year*
Memo: ROIC on GAAP EBIT (litigation left in)		41.0%	46.3%	49.1%	46.6%	54.2%	58.4%	63.6%	68.7%	74.1%	*Shows the litigation drag on reported returns*
Operating ROIC — excluding goodwill (return on the operating network)											
Invested capital ex-goodwill — ending	22,391	19,586	19,150	23,312	23,205	25,613	25,613	25,613	25,613	25,613	*Link: Tab 2 (Invested capital – Goodwill)*
Invested capital ex-goodwill — average		20,989	19,368	21,231	23,259	24,409	25,613	25,613	25,613	25,613	*Formula: (Opening + Closing) / 2*
Operating ROIC — on average capital		**77.3%**	**93.0%**	**93.5%**	**94.7%**	**103.6%**	**108.1%**	**119.6%**	**131.2%**	**143.6%**	*Formula: NOPAT / average invested capital ex-goodwill*
Memo: Tangible invested capital (ex-goodwill & intangibles)	(5,273)	(5,479)	(6,954)	(3,577)	(4,441)	(2,033)	(2,033)	(2,033)	(2,033)	(2,033)	*Negative every year → ROIC on tangible capital is not meaningful (see note 1)*
The economic spread & value creation											
WACC (today's, applied to all years)	9.7%	9.7%	9.7%	9.7%	9.7%	9.7%	9.7%	9.7%	9.7%	9.7%	*Link: WACC cell above (see note 2)*
ROIC spread		**33.2%**	**38.6%**	**40.3%**	**41.9%**	**46.9%**	**49.6%**	**54.8%**	**59.9%**	**65.3%**	*Formula: ROIC (average capital) – WACC*
Operating ROIC spread		**67.7%**	**83.3%**	**83.9%**	**85.0%**	**93.9%**	**98.4%**	**109.9%**	**121.5%**	**134.0%**	*Formula: Operating ROIC – WACC*
Economic profit ($mm)		**12,568**	**14,399**	**16,019**	**17,888**	**20,964**	**23,168**	**26,031**	**28,936**	**32,038**	*Formula: NOPAT – WACC × average invested capital (= spread × capital)*
Capital charge ($mm)		3,663	3,605	3,841	4,128	4,323	4,517	4,595	4,672	4,749	*Formula: WACC × average invested capital — the 'rent' on investors' money*

Economic profit as % of NOPAT	77.4%	80.0%	80.7%	81.3%	82.9%	83.7%	85.0%	86.1%	87.1%	*Share of after-tax operating profit that is pure value creation*

WACC sensitivity — beta (rows) × equity risk premium (columns); capital weights and Kd held constant

Beta \ ERP	**4.0%**	**4.5%**	**5.0%**	**5.5%**	**6.0%**	*Beta used in CAPM (after any Blume adjustment): 0.59 = low · 0.77 · 0.85 ≈ base · 0.95 · 1.10 = stress*
0.59 β	7.5%	7.7%	8.0%	8.3%	8.6%	*Even the stress case (≈11.6%) leaves FY25 ROIC (51.6%) about 40 points above WACC.*
0.77 β	8.2%	8.5%	8.9%	9.3%	9.6%	
0.85 β	8.5%	8.9%	9.3%	9.7%	10.1%	
0.95 β	8.8%	9.3%	9.8%	10.2%	10.7%	
1.10 β	9.4%	10.0%	10.5%	11.0%	11.6%	

FY25 economic profit sensitivity ($mm) — same axes · = FY25 NOPAT – WACC × FY25 average invested capital

Beta \ ERP	**4.0%**	**4.5%**	**5.0%**	**5.5%**	**6.0%**	*Formula: NOPAT – (WACC from the grid above) × average invested capital. Live: updates with every input.*
0.59 β	18,836	18,714	18,592	18,470	18,348	*Range ≈ $17.1B–$18.8B, i.e. 78–86% of FY25 NOPAT ($22.0B) is value created after the capital charge in every case.*
0.77 β	18,538	18,379	18,220	18,061	17,902	
0.85 β	18,406	18,230	18,055	17,879	17,703	
0.95 β	18,241	18,044	17,848	17,652	17,455	
1.10 β	17,993	17,765	17,538	17,311	17,083	

Notes

1. Tangible invested capital is negative in every year (FY25: –$4.4B): the operating network is funded by operating liabilities (client-incentive accruals, settlement obligations, deferred taxes), not shareholder capital. The return on tangible capital is therefore not meaningful — more precisely, Visa needs no net tangible capital to grow. Operating ROIC excludes goodwill only.

2. WACC uses today's market inputs (10-year yield 5.19%) for every historical year — a simplification. Rates were much lower in FY21–FY22, so historical spreads are, if anything, understated.

3. ROIC uses EBIT ex-litigation. On GAAP EBIT, FY25 ROIC falls by ~5 points, but the spread is still above 35 points.

4. Cost of debt is a marginal market rate (traded yield on Visa's 2035 notes), not Visa's ~3% effective rate on legacy low-coupon notes. Debt is ~3% of capital, so this choice moves WACC by only a few basis points. Limitation: the implied ~23 bp spread over Treasuries is tight for AA- nine-year paper and the bond quote is undated; even +50 bp on the cost of debt moves WACC by ~1 bp.

5. Projections (FY26P–FY30P) link to Tab 2. ROIC keeps rising (to ~75% by FY30P) because NOPAT grows ~9–10% a year while invested capital grows only ~1–2% (acquisitions only — Visa returns ~95% of FCF). That is the toll-road mechanic in one line, but treat the outer years as an upper bound: share-based comp and any larger acquisition would add capital. Operating ROIC ex-goodwill exceeds 100% because acquisitions are booked to goodwill.

Tab 5 — Peer Spread: Visa vs Mastercard

Mastercard rebuilt with the SAME definitions as Tabs 1–4 · $ in millions · MA fiscal year = calendar year (3-month offset to Visa) · green = link · black = formula

Mastercard — line item	FY21	FY22	FY23	FY24	FY25	Formula / note
Volume & revenue						
Purchase volume ($ billions)	5,974	6,571	7,345	8,003	8,799	*Comparable to Visa payments volume (both exclude cash)*
Net revenue	**18,884**	**22,237**	**25,098**	**28,167**	**32,791**	
Net revenue growth		17.8%	12.9%	12.2%	16.4%	
Take rate (bps of purchase volume)	**31.6 bps**	**33.8 bps**	**34.2 bps**	**35.2 bps**	**37.3 bps**	*Formula: Net revenue / purchase volume · same construction as Visa's core take rate*
Profitability						
Operating income (EBIT) — GAAP	10,082	12,264	14,008	15,582	18,897	
Operating margin — GAAP	53.4%	55.2%	55.8%	55.3%	57.6%	
Add back: Provision for litigation	94	356	539	680	504	
EBIT ex-litigation	**10,176**	**12,620**	**14,547**	**16,262**	**19,401**	*Same treatment as Visa (Tab 2)*
Operating margin — ex-litigation	53.9%	56.8%	58.0%	57.7%	59.2%	
Tax rate — GAAP effective	15.7%	15.4%	17.9%	15.6%	19.4%	*FY25 19.4% vs 15.6% in FY24 (4Q25 release)*
Net income	8,687	9,930	11,195	12,874	14,968	
Net margin	46.0%	44.7%	44.6%	45.7%	45.6%	
Returns on capital						
NOPAT	**8,577**	**10,682**	**11,940**	**13,725**	**15,631**	*Formula: EBIT ex-litigation × (1 – tax rate)*
Total debt	13,901	14,023	15,681	18,226	19,000	
Total equity	7,383	6,356	6,975	6,515	7,746	
Cash + investments deducted	7,894	7,408	9,180	8,772	10,898	*Restricted litigation cash not deducted (as for Visa's escrow)*
Invested capital	**13,390**	**12,971**	**13,476**	**15,969**	**15,848**	*Formula: Debt + Equity – Cash (Tab 2 definition)*
Average invested capital		13,181	13,224	14,723	15,909	
ROIC — on average capital		**81.0%**	**90.3%**	**93.2%**	**98.3%**	
Goodwill	7,662	7,522	7,660	9,193	9,560	
Other intangible assets, net	3,671	3,859	4,086	5,453	5,554	
Invested capital ex-goodwill	5,728	5,449	5,816	6,776	6,288	
Operating ROIC — ex-goodwill (average)		**191.1%**	**212.0%**	**218.0%**	**239.3%**	
Memo: Tangible invested capital	2,057	1,590	1,730	1,323	734	*Small but positive for MA; negative for Visa*
Cash flow & capital allocation						
CapEx (PP&E + capitalized software)	814	1,097	1,088	1,194	1,215	*Both lines, to match Visa's 'property, equipment and technology'*
Free cash flow	**8,649**	**10,098**	**10,892**	**13,586**	**16,433**	*Formula: OCF – CapEx*
FCF conversion (FCF / net income)	99.6%	101.7%	97.3%	105.5%	109.8%	
Capital intensity (CapEx / net revenue)	4.3%	4.9%	4.3%	4.2%	3.7%	
Capital returned (dividends + buybacks)	7,645	10,656	11,190	13,402	14,483	
Capital returned as % of FCF	88.4%	105.5%	102.7%	98.6%	88.1%	
Dividend payout ratio	20.0%	19.2%	19.3%	19.0%	18.4%	
Acquisitions (cash)	4,436	313	-	2,511	-	*Lumpy: FY21 $4.4B and FY24 $2.5B (e.g. Recorded Future)*

Mastercard WACC (same market inputs as Tab 3)		
Risk-free rate	5.2%	*Link: Tab 3*
Equity risk premium	5.5%	*Link: Tab 3*
Raw beta	0.76	*Data_MA (Yahoo 5Y monthly, third-party)*
Adjusted beta	0.84	*Blume-adjusted, as for Visa*
Cost of equity	**9.8%**	*Rf + β × ERP*
After-tax cost of debt	4.4%	*Assumes MA borrows at Visa's market yield (both high-grade); × (1 – MA FY25 tax rate)*
Market capitalisation	528,438	*Price × shares (Data_MA)*
Total debt (Jun 30, 2026)	24,643	*Data_MA*
Equity weight	95.5%	
Mastercard WACC	**9.6%**	*Formula: E-weight × Ke + D-weight × Kd*

FY25 head-to-head	Visa	Mastercard	Visa – MA	Why the difference matters
Net revenue ($mm)	40,000	32,791	7,209	*Visa is ~1.2x Mastercard's revenue*
Net revenue CAGR FY21–FY25	13.5%	14.8%	-1.3 pts	*MA grew faster, helped by value-added services and cross-border recovery*
Volume ($ billions): Visa payments vs MA purchase	14,217	8,799	5,418	*Visa's network is ~1.6x larger by volume*
Take rate (bps)	28.1 bps	37.3 bps	-9.1 bps	*MA earns MORE per dollar: higher value-added-services mix and more cross-border; Visa's scale is in debit*
Operating margin — GAAP	60.0%	57.6%	+2.4 pts	
Operating margin — ex-litigation	66.4%	59.2%	+7.2 pts	*Visa is ~7 pts more profitable per revenue dollar — scale economics of the larger network*
Net margin	50.1%	45.6%	+4.5 pts	
Invested capital ($mm)	43,084	15,848	27,236	*Visa carries ~2.7x the capital — mostly Visa Europe goodwill & intangibles*
Goodwill + intangibles ($mm)	47,525	15,114	32,411	*$47.5B vs $15.1B: the 2016 Visa Europe acquisition (~$23B) is the single biggest gap*
ROIC — average capital	51.6%	98.3%	-46.7 pts	*MA's ROIC is ~2x Visa's. The gap is goodwill & intangibles ($47.5B vs $15.1B). Debt-funded buybacks do NOT explain it: they swap equity for debt, leaving debt + equity – cash unchanged*
Operating ROIC — ex-goodwill	94.7%	239.3%	-144.6 pts	*Both are effectively capital-free networks once acquisition goodwill is removed*
Tangible invested capital ($mm)	(4,441)	734	(5,175)	*Both near zero or negative — neither network needs net tangible capital*
WACC	9.7%	9.6%	+0.1 pts	*Near-identical: same market inputs and almost the same beta (0.77 vs 0.76)*
ROIC – WACC spread	41.9%	88.7%	-46.8 pts	*Both spreads are enormous; the gap is about capital base, not moat quality*
Economic profit ($mm)	17,888	14,110	3,779	*Visa still creates MORE absolute value: bigger profit pool outweighs its larger capital charge*
FCF conversion	107.6%	109.8%	-2.2 pts	
Capital intensity	3.7%	3.7%	-0.0 pts	*Both under 4% of revenue*
Capital returned, % of FCF (FY21–25 cumulative)	93.1%	96.2%	-3.1 pts	*Both return ~all FCF; MA's equity is only ~$7.7B because buybacks have run ahead of retained earnings for years*
Dividend payout ratio	23.1%	18.4%	+4.7 pts	*Similar ~20% dividend policies; buybacks do the rest*
Market cap today ($mm)	735,888	528,438	207,450	*Both at the Oct 14, 2026 close*
P/E on FY25 net income	36.7x	35.3x	1.4x	*The market prices the two almost identically despite MA's 2x ROIC — consistent with the ROIC gap being acquisition accounting, not economics. (MA FY25 = calendar 2025, 3 months more recent)*

Notes

1. Same definitions as Visa: EBIT ex-litigation, GAAP effective tax, invested capital = debt + equity – (cash + investments), restricted litigation cash not deducted, CapEx includes capitalised software.

2. Timing: Mastercard FY25 = calendar 2025; Visa FY25 = Oct 2024–Sep 2025. A 3-month offset — immaterial for ratio comparisons, but it means Mastercard's FY25 earnings base is a quarter more recent.

3. The headline: Mastercard's ROIC is roughly double Visa's, but this is an acquisition-accounting story, not a moat-quality story. (ROIC including goodwill remains the right measure of the return on the ~$23B Visa actually paid for Visa Europe — the goodwill is real money, not an artifact.) Visa is more profitable per revenue dollar and generates more economic profit in absolute dollars; its ROIC is diluted by ~$47B of Visa Europe goodwill and intangibles. Strip goodwill out and both networks need essentially no tangible capital.

4. Mastercard's low equity (~$7.7B) reflects years of debt-funded buybacks — it inflates ROE far more than ROIC, which is why ROIC (debt + equity) is the fairer comparison.

5. Mastercard's beta (0.76) is a third-party Yahoo figure, as is Visa's. It affects only MA's WACC, not the ROIC comparison.